WebGPU Geant4-DNA: browser-native, kernel-fused Monte Carlo track-structure simulation for radiobiology, validated against Geant4 11.4.1

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Abstract

We present a WebGPU/WGSL re-implementation of Geant4-DNA track-structure physics and Karamitros Independent-Reaction-Time (IRT) radiolysis chemistry that runs entirely in a web browser, with no CUDA toolchain and no installation. The simulation uses a *kernel-fusion* architecture: one GPU thread per primary electron executes the full particle history inside a single fused compute dispatch, eliminating per-step dispatch overhead. We do not transpile Geant4; we re-implement its physics models from the Geant4-DNA source and the G4EMLOW 8.8 tabulated cross sections, and use a normally-compiled Geant4 11.4.1 as a validation oracle on a falsifiable, artifact-backed ladder. Cross sections match the reference to peak ratios of 0.975 to 1.000; the continuous-slowing-down range at 10 keV is 0.988× Geant4. WebGPU track-structure tracking (Phases A+B) runs **455**× faster than Geant4 single-thread (280× vs. an 8-thread build) on an Apple M2 Pro, but we are explicit about scope: the Geant4 figure is whole-process wall-clock and the WGSL figure is dispatch-only. We measured the gap — Geant4 init plus DNA table-build is a negligible ∼2 s (so event-loop-only the figure is 452×), the one real asymmetry being the ∼6.8 GB of per-event ntuple I/O Geant4 writes and WGSL does not — and we note that the kernel-fusion architecture itself accounts for only ∼2× of the speedup (the fused Phase A is 2 % of the tracking time; Phase B is an un-fused wavefront). The honest like-for-like figure is the *end-to-end* **1.48**×, where both pipelines do the same work and the IRT radiolysis chemistry on CPU dominates the wall-clock—we report all of these and are explicit about which is which. We also report the project's honest limits: a 27 % cascade-ionisation deficit traceable to an intentional excitation cross-section choice, and the finding that the *inter-track* chemistry contribution—which closes most of the residual 1 µs gap against Geant4's `chem6`—is *not* reachable browser-native, because cross-primary IRT is $O(N^2)$ for a point source and GPU step-by-step chemistry is bounded by the encounter radius to ∼50 000 time steps. These negative results are quantified and motivate a native runtime as the path forward. All experiments are reproducible via committed JSON artifacts.

1 Introduction

Geant4-DNA [9, 2, 10] is the CNRS/IN2P3-coordinated extension of Geant4 [1] for discrete, event-by-event Monte Carlo track-structure simulation in liquid water at radiobiologically relevant (eV–keV) energies, together with the subsequent radiolysis chemistry. It is the de-facto reference for mechanistic modelling of radiation-induced DNA damage.

GPU acceleration of this class of simulation is an active area: gMicroMC [14] and MPEXS-DNA [13] target track-structure on the GPU, while GGEMS [3] and GPUMCD [7] target macroscopic dose. All of these are built on CUDA or OpenCL and require a workstation-class install. To our knowledge there is no open-source GPU Monte Carlo for radiation transport that runs *browser-native*—i.e. on the cross-vendor WebGPU API, with zero install, on any machine with a modern browser. That accessibility gap is the motivation here: a browser-native engine is trivially shareable, reproducible without a toolchain, and usable for teaching and rapid validation.

Our contributions are: (i) a complete WebGPU/WGSL re-implementation of the Geant4-DNA DNA_Opt2 physics chain plus Karamitros IRT chemistry; (ii) a kernel-fusion architecture that yields large speedups over single- and multi-threaded Geant4; (iii) a research-grade, falsifiable validation ladder against compiled Geant4 11.4.1, with every claim backed by a committed artifact; and (iv) an honest characterisation of where the browser-native approach reaches its limits, including a quantified negative result on the inter-track chemistry contribution.

2 Methods

2.1 Re-implementation, not transpilation

None of Geant4's runtime executes on the GPU. Each physics model is re-implemented as flat WGSL arithmetic, using the Geant4-DNA C++ source as the algorithmic specification (sampling routines and formulae) and the G4EMLOW 8.8 data files as the cross-section tables, converted once into WGSL constant arrays. A normally-compiled Geant4 11.4.1 / G4EMLOW 8.8 install serves as the *validation oracle*: we run the `dnaphysics` example single-threaded, dump its ntuples, and compare the WGSL output against them—the same validate-against-compiled-Geant4 methodology used by gMicroMC, MPEXS-DNA and GGEMS.

2.2 Kernel-fusion architecture

The pipeline has three phases (Fig. 1): **Phase A (primaries)** is a single fused `@compute` dispatch with one thread per primary electron; that thread runs the entire history—Born ionisation, Emfietzoglou excitation, elastic and vibrational scattering—in a `for` loop, with no per-step dispatch. **Phase B (secondaries)** is a wavefront stepper (one dispatch per physics step advancing all live secondaries), which cannot be fused because the secondary count is unknown until Phase A completes. **Phase C (chemistry)** runs the Karamitros 9-reaction IRT in a Web Worker off the main thread (a GPU spatial-hash alternative exists but is less accurate at long times; see Sec. 4). Energy deposition into a 128^3 voxel grid uses fixed-point integer `atomicAdd` ($\times100$ per eV), since WGSL atomics are integer-only.

2.3 Ported models and deliberate divergences

We port Born ionisation (5 shells), Champion tabulated / screened-Rutherford elastic, Sanche vibrational excitation (9 modes), and the Karamitros 9-reaction IRT chemistry (6 species, Smoluchowski totally-diffusion-controlled and Onsager-screened partially-diffusion-controlled types) [11, 12]. Two deliberate departures from the `DNA_Opt2` defaults are documented (Table 1): we use the Emfietzoglou excitation model [5]—the model used by the Geant4-DNA constructor `G4EmDNAPhysics_option4`, which is unrelated to the similarly-named EM Standard list `G4EmStandardPhysics_option4`—rather than Born, because it yields the correct initial $G(\mathrm{H}) = 0.33$; and one tuning scalar, `SIGMA_EXC_SCALE`$= 0.5$, scaling the excitation cross section. A second scalar present in the codebase, `RECOMB_BOOST`, has no physical basis; we measured it to be nearly inconsequential (Sec. 5, E10r) and set it to its neutral value 1.0 for all reported chemistry, so the G-values of §3 are parameter-free in that knob.

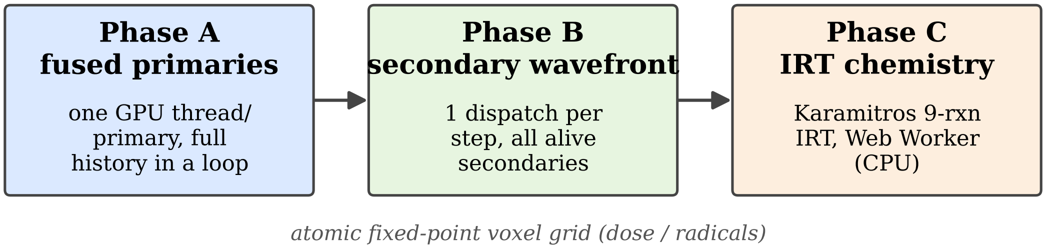

Figure 1: Three-phase pipeline. Phase A is fully fused (one GPU thread per primary, full history in a loop); B and C cannot be.

Table 1: Deliberate divergences from Geant4-DNA `DNA_Opt2`, with the measured cost of each (experiment IDs reference § Numbers).

Item	`DNA_Opt2`	This work	Cost
Excitation	Born	Emfietzoglou	σ_{exc} 2.55× (E6b)
σ_{exc} scale	—	×0.5	net 1.27× (E6c)
Recomb. boost	—	off (×1.0)	Δ1.4 pp RMS (E10r)
Chemistry pool	single	per-primary	$\Delta G(\mathrm{H_2})$ (E10f)
Dose reduction	fp64	fp32 atomics	MC-equivalent, not bit-exact

2.4 WGSL constraints

WGSL forbids recursion, dynamic dispatch and dynamic allocation, and allows atomics only on `u32`. Consequently: all physics is inlined into `main()`; secondaries are deferred to a buffer + wavefront rather than recursed; dose uses fixed-point integers; and all memory is pre-allocated.

2.5 Validation protocol

The repository follows a six-level falsifiable ladder. Every experiment emits a JSON artifact carrying the git SHA, a named seed, WGSL shader hashes, a pass bar, and per-row observations; failed experiments are committed with `status:"fail"` and a diagnosis rather than rerun. All numbers below are drawn from this artifact set. The reference configuration is $N = 4096$ primaries at 10 keV on an Apple M2 Pro (`apple/metal-3` adapter). A reproducibility caveat applies: fp32 atomic reductions are not order-deterministic across GPU vendors, so cross-hardware results are statistically equivalent within Monte Carlo noise but not bit-exact.

3 Results

3.1 Cross sections (Level 1)

All 9 cross-section channels match the G4EMLOW reference (Table 2). Sanche vibrational is bit-exact; the largest deviation is the Champion elastic peak ratio at 0.975.

Table 2: Cross-section validation vs. G4EMLOW 8.8. Peak ratio is WebGPU/Geant4 at the worst single point; medians of the relative deviation are $\sim 10^{-3}$.

Channel	Model	Peak ratio	Exp.
Ionisation σ_{ion}	Born (5 shells)	0.9987	E1
Excitation σ_{exc}	Emfietzoglou (5 lev.)	0.9970	E2
Elastic σ_{el}	Champion / scr.-Ruth.	0.9751	E3
Vibrational σ_{vib}	Sanche (9 modes)	1.0000	E4

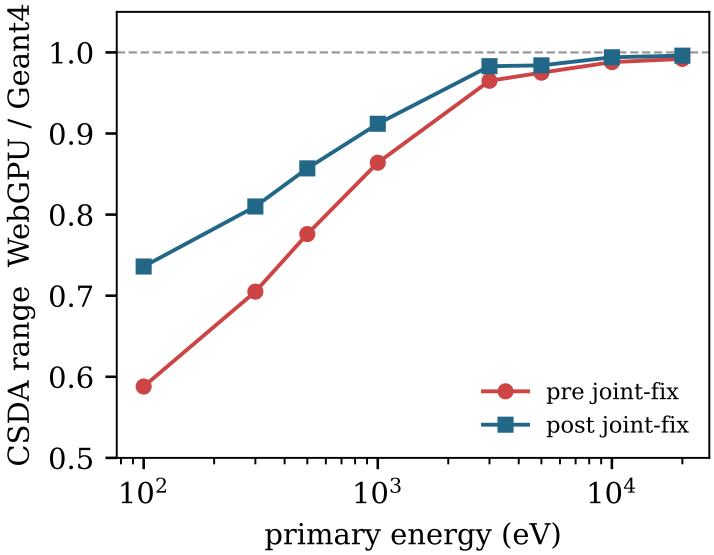

Figure 2: Energy-dependent CSDA closure under the joint fix: WebGPU/Geant4 range ratio at the eight ESTAR energies, pre vs. post fix (8/8 improved; artifacts E5b, E5d).

3.2 Track structure (Level 2)

The continuous-slowing-down (CSDA) range at 10 keV is 2714.4 nm vs. Geant4's 2747.5 nm, i.e. 0.988× (3.59σ), with energy conservation at 100.0 % (E5). Across the eight ESTAR energies the deficit is energy-dependent and was closed monotonically by the joint fix: the 100 eV ratio rose from 0.587× to 0.736× and all eight energies improved, with the lift inversely proportional to the original deficit—the expected signature of the excitation cross-section correction (E5b, E5d; Fig. 2).

The mean free path is 5–11 % short of Geant4 across six energy bins (median 0.941, E6). The full-cascade ionisation count is 371.9 per primary vs. Geant4's 509.2—a 27 % deficit at 263σ (E7)—and the implicit W-value is 26.89 eV vs. ICRU 31's 21.4 eV ([8], +25.7 %, E5c). Both are the same physics: the intentional excitation inflation channels energy away from ionisation. The secondary-electron kinetic-energy spectrum at creation matches to 1.0 % overall, with 7/8 log-bins within 0.1–3.1 % (E8). Table 3 summarises.

3.3 Chemistry (Levels 3–4)

We report *parameter-free* chemistry yields: the empirical `RECOMB_BOOST` knob (Table 1), which has no first-principles basis, is set to its neutral value 1.0. Against Geant4's `chem6` at matched 10 keV, the G-values at 1 µs (E10r) are then $G(\mathrm{OH})$ 0.91×, $G(\mathrm{e}^-_{\mathrm{aq}})$ 0.86×, $G(\mathrm{H})$ 0.93×, $G(\mathrm{H}_2)$

Table 3: Track-structure metrics vs. Geant4 11.4.1 at 10 keV, $N = 4096$.

Metric	This work	Geant4	Ratio (exp.)
CSDA range	2714.4 nm	2747.5 nm	0.988 (E5)
Energy cons.	100.0 %	99.99 %	1.000 (E5)
Ions / primary	371.9	509.2	0.730 (E7)
W-value	26.89 eV	21.4 eV†	1.257 (E5c)
Sec. KE/primary	143.4	144.9	1.010 (E8)

† ICRU 31 low-LET liquid water reference.

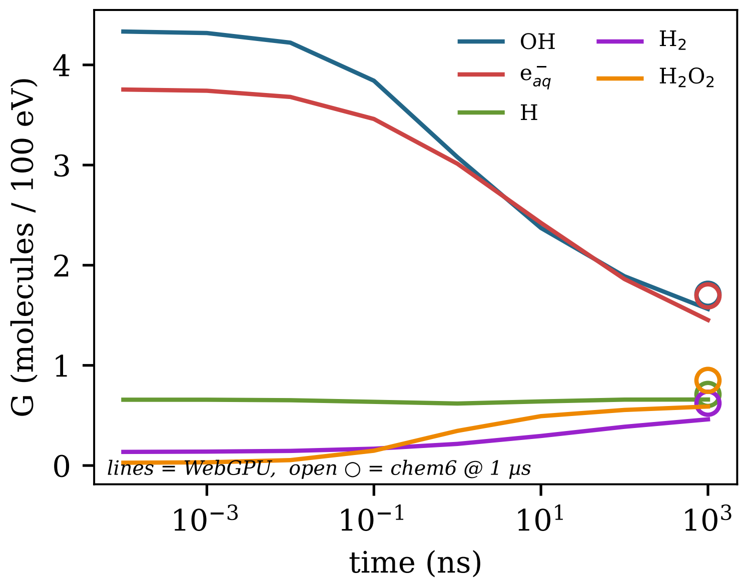

Figure 3: Radiolysis G-value time evolution—parameter-free WebGPU yields (`RECOMB_BOOST` = 1.0; lines) against Geant4 `chem6` endpoints at 1 µs (open circles). Artifact E10r.

0.74×, $G(H_2O_2)$ 0.69× (Fig. 3), a five-species RMS deviation of 19.7 % at 1 µs.

Crucially, we measured that the knob is *not* load-bearing: enabling it (×2.0) changes the overall RMS deviation by only ∼1.4 pp (to 18.3 %), and although it lifts $G(H_2)$ it does so by introducing a 1.10× overshoot in G(H) while leaving $G(H_2O_2)$ unchanged—so we drop it and report the parameter-free values above (E10r). A first-principles replacement for its intended effect (super-excitation autoionisation, or an Onsager/Mozumder escape-probability treatment of geminate recombination) is left to future work (Sec. 5).

One result is energy-resolved and tuning-independent: $G(e^-_{aq})$ is non-monotonic between 1 and 3 keV, a 12.5 % dip significant at 126σ by primary-bootstrap (Fig. 4, E10b), which `chem6` independently reproduces (E10d)—real track-end / spur-structure physics, not an artifact.

3.4 DNA damage (Level 5)

We report Level 5 as a *calibrated* damage model, not a validated prediction, and state the distinction plainly. On a 21 × 21 B-DNA fibre grid (3.89 Mbp target) the strand-break counts are the geometric backbone-reach counts multiplied by per-event break probabilities: $\mathrm{SSB_{dir}} = 26 = \lfloor 173 \times P_{\mathrm{dir}} \rfloor$ with $P_{\mathrm{dir}} = 0.15$, and $\mathrm{SSB_{ind}} = 64 \approx 1423 \times P_{\mathrm{ind}}$ with $P_{\mathrm{ind}} = 0.05$. The reported indirect/direct ratio

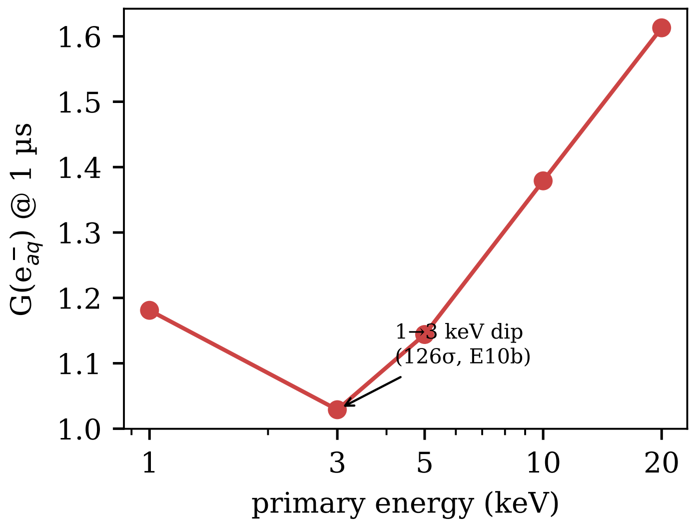

Figure 4: $G(e^-_{aq})$ at 1 µs vs. primary energy: the non-monotonic 1→3 keV dip (126σ, E10b), independently reproduced by `chem6` (E10d).

of 2.46 lies within PARTRAC's 2–3 band [6], but P_{ind} was *calibrated* (from the Geant4 default 0.4) to land in that band; the ratio is therefore a fit, not an independent test, and PARTRAC is itself a simulation.

Against experiment-calibrated cellular yields (∼35 DSB and ∼1000 SSB per diploid cell per Gy for low-LET radiation [15]), the absolute per-Da yields are high by 223× (SSB) and 796× (DSB) *when normalised by the box-average dose* (0.243 Gy). This is entirely a point-source dose-normalisation artifact, which we quantify (E12-local): the validation dumps launch all primaries from the origin, so 98.1 % of the deposited energy — measured from the pre-chemistry radical-position dump — lands in the central 3 µm fibre-core cube. The dose the DNA actually receives is therefore ∼238 Gy (concentration factor $C \approx 981$), not 0.243 Gy. Re-normalised by this local dose, the absolute yields are 0.34× **(direct SSB),** 0.82× **(DSB), and** 1.28× **(total SSB)** of experiment — within a factor of ∼3, not orders of magnitude. The one residual is the DSB/SSB *ratio* (3.6× high), which dose normalisation does not touch and which traces to the calibrated P_{ind} above. Level 5 thus predicts absolute DNA-damage yields to order-of-magnitude once the dose is scored where the DNA sits; the strand-coincidence ratio remains a calibration, not a prediction. (A cleaner follow-up, E12-bulk, would distribute the primary start positions across the box so that box-average and local dose coincide and the C correction is unnecessary.)

3.5 Performance (Level 6)

Phase A+B tracking is **455**× faster than Geant4 11.4.1 single-thread (635 ms vs. a 289.1 s median over 3 trials, E15b) and **280**× vs. an 8-thread build (178.0 s, E15c). Peak Phase A throughput is 538 947 primaries/s at $N = 16384$ (E15). We state two caveats so these numbers are not over-read. *First, on apple-to-apple scope*: the Geant4 wall-clock is whole-process while the WGSL 635 ms is dispatch-only, but we measured the difference and it is small where it matters. A 16-primary init-probe runs in 3.2 s (E15-fair), so Geant4 process init plus DNA physics-table construction is only ∼2.1 s(0.7 % of 289 s); the 289 s is ∼99 % genuine event-loop, and event-loop-only the speedup is 452×, statistically identical to 455×. The one material asymmetry is per-event ntuple I/O —

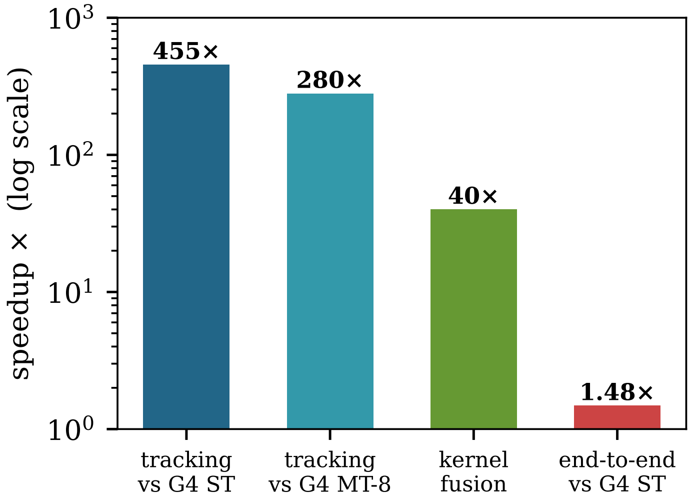

Figure 5: Speedups (log scale): fused track-structure kernel vs. Geant4 single-thread (E15b) and 8-thread (E15c), the within-WebGPU kernel-fusion factor (E16), and the honest *end-to-end* figure (E15b), which is only 1.48× because the IRT chemistry runs on CPU.

Geant4 writes ∼6.8 GB to disk over the full run (the row-wise merge is the likely cause of the poor 1.62× MT-8 scaling), which the WGSL dispatch does not; a no-ntuple Geant4 build would lower its number somewhat, so 452× is a mild over-estimate. (We note for transparency that an earlier draft mis-estimated the fixed overhead at ∼160 s from the MT scaling alone; the direct measurement corrects this.) *Second, kernel fusion is not what delivers the large factor*: a within-WebGPU comparison of the fused kernel against a modelled per-step dispatch gives a **40**× factor *for Phase A* (E16), but Phase A is only 14.4 ms (2 %) of the 635 ms; Phase B is an un-fused 2000-dispatch wavefront. Fusion's contribution to the full tracking pipeline is therefore ∼2× (1324 ms naive → 635 ms fused), with GPU data-parallelism over N primaries supplying the rest. The single honest like-for-like number is the *end-to-end* 1.48×, where both pipelines do the same work and the IRT chemistry on CPU dominates the wall-clock (194 s of 194.6 s); this motivates Sec. 4.

4 The inter-track chemistry limitation

The largest remaining gap vs. `chem6` is an *inter-track* effect: per-primary IRT partitioning misses cross-primary reactions that, when included, add $\Delta G(\mathrm{H}_2) = +0.149$ at 1 µs—96 % of the residual implementation gap (E10f). We asked whether this is recoverable browser-native, and found it is not, by two independent routes.

Cross-primary IRT is $O(N^2)$ for a point source. The IRT reaction horizon is the diffusion length over the full 1 µs window, $R_{\mathrm{cut}} = 1.45 + 2\sqrt{8D_{\max}t_{\max}} \approx 551\,\mathrm{nm}$, not the few-nm contact scale. At 10 keV the radicals occupy a ∼5 µm blob, so a 551 nm spatial hash spans most of it and prunes almost nothing; over 50 % of accepted cross-primary reactions involve pairs more than 100 nm apart (E10k). The single-pool scan is $\sim 10^{12}$ reaction-time samples (∼70 hr in-browser) and the heap is ∼2 GB.

GPU step-by-step (SBS) chemistry is encounter-radius-bounded. The GPU radiolysis codes use a step-by-step Brownian-motion model in which a reaction fires when two reactants fall within a Smoluchowski reaction radius (gMicroMC [14]; MPEXS-DNA adds a during-step

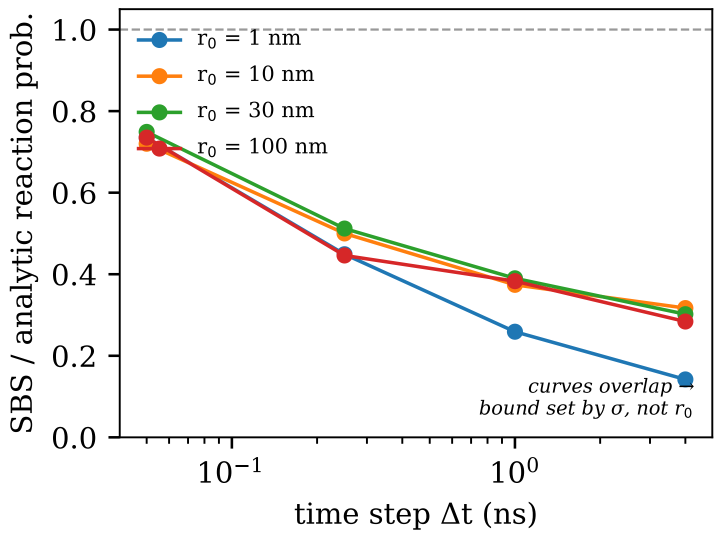

Figure 6: Kill criterion (E10Q): single-pair SBS/analytic reaction-probability ratio vs. Δt for four separations. The curves overlap, so the Δt requirement is set by the encounter radius σ, not by separation — far, dilute pairs need the same tiny steps as close ones.

reaction probability at small time steps [13]). We implemented this in WGSL and, as a refinement, a Green's-function (Brownian-bridge) during-step reaction probability [4] from the step-by-step radiation-chemistry literature, and measured its convergence against the analytic Smoluchowski value for a single pair. The per-step cost is $\sim$110 ms at $N = 4096$ (E10L), and the IRT-parity budget is $\sim$1750 steps. But accuracy requires a time step that resolves the encounter radius σ, $\Delta t \lesssim \sigma^2/(2D) \approx 0.02\,\mathrm{ns}$, *independent of pair separation or density* (E10Q; Fig. 6)—so $\sim$50 000 steps are needed regardless. Compaction lowers per-step cost but not step count (E10P). No SBS schedule, full or hybrid, beats the IRT worker at the required accuracy on laptop WebGPU.

Both routes converge on the same conclusion: the inter-track physics requires a *native* runtime (e.g. Node/Deno + `wgpu-native`), where RAM is cheap (unblocking single-pool cross-primary IRT) or many cores parallelise the step count. The browser remains the validation and demonstration surface.

5 Discussion

Browser-native execution buys accessibility and reproducibility that the CUDA/OpenCL engines cannot: the simulation and its validation harness run on any modern GPU with no install, which is valuable for teaching, sharing, and independent re-checking of every artifact. The cost is the WebGPU feature envelope (integer-only atomics, no recursion) and—per Sec. 4—an inter-track chemistry ceiling that pushes the last increment of fidelity off-browser. The cascade-ionisation deficit and the `RECOMB_BOOST` fudge are open items with identified, falsifiable next steps (super-excitation autoionisation; an Onsager escape-probability treatment to replace the fudge). Relative to the CUDA codes gMicroMC and MPEXS-DNA, our speedups come from kernel fusion on a single consumer GPU (MPEXS-DNA, for reference, reports up to 2900× over single-core Geant4-DNA on an NVIDIA TITAN V [13]); extending the same methodology to the EM Standard list `G4EmStandardPhysics_option4` for macroscopic photon dose is a natural, if substantially larger, direction.

6 Conclusion

A browser-native, kernel-fused WebGPU re-implementation reproduces Geant4-DNA track structure to within a fraction of a percent on cross sections and 0.988× on CSDA range, at 455×/280× the speed of single-/multi-threaded Geant4, with an honest, artifact-backed account of its limits—most notably the quantified result that the inter-track chemistry contribution is not browser-native and requires a native runtime.

Data and code availability

Source, the full validation ladder, and every JSON artifact are at `https://github.com/abgnydn/webgpu-dna`; a live build runs at `https://webgpudna.com`. A versioned Zenodo deposit accompanies release `v0.4.0` (DOI to be inserted). Each experiment reruns via `npm run experiments -<id>`.

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